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United States Patent	12415326
Kind Code	B2
Date of Patent	September 16, 2025
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Puncture repair liquid unit and puncture repair kit

Abstract

A puncture repair liquid unit and a puncture repair kit each has an extraction cap and a container. The extraction cap has a first flow path, a second flow path, and a third flow path. The first flow path extends between a first inlet connectable to an external compressor device for generating compressed air and a first outlet opening in the container. The second flow path extends between a second inlet opening in the container and a second outlet for taking out a puncture repair liquid from the extraction cap to supply to a punctured tire to be repaired. The second inlet communicates with the first outlet via the space of the container. The third flow path communicates with the first and second flow paths outside the space of the container so that a portion of the compressed air in the first flow path flows into the second flow path.

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Appl. No.: **17/961719**

Filed: **October 07, 2022**

Prior Publication Data

Document Identifier	Publication Date
US 20230140616 A1	May. 04, 2023

Foreign Application Priority Data

JP	2021-176795	Oct. 28, 2021

Publication Classification

Int. Cl.: **B29C73/02** (20060101)

U.S. Cl.:

CPC **B29C73/025** (20130101);

Field of Classification Search

CPC: B29C (73/025)

USPC: 141/37

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Background/Summary

RELATED APPLICATIONS

(1) This application claims the benefit of foreign priority to Japanese Patent Applications No. JP2021-176795, filed Oct. 28, 2021, which are incorporated by reference in its entirety.

FIELD OF THE INVENTION

(2) The present disclosure relates to a puncture repair liquid unit and a puncture repair kit.

BACKGROUND OF THE INVENTION

(3) Japanese Unexamined Patent Application Publication No. 2018-192667 (Patent Document 1) has disclosed a bottle unit for puncture repair. This bottle unit has a bottle container containing puncture repair liquid and a cap attached to the mouth of the bottle container.

(4) The cap has a first flow path and a second flow path. The first flow path has a first opening portion open in the bottle container to feed compressed air from a compressor into the bottle container. The second flow path has a second opening portion open in the bottle container to sequentially taking out the puncture repair liquid and the compressed air from the bottle container by feeding compressed air.

SUMMARY OF THE INVENTION

(5) However, with the bottle unit described above, a portion of the puncture repair liquid tends to harden near the valve of the punctured tire to be repaired, and the compressed air supply path is partially blocked, which causes a problem of requiring a lot of time to supply compressed air to the tire.

(6) The present disclosure was made in view of the above, and a primary object thereof is to provide a puncture repair liquid unit capable of reducing the time required to supply compressed air to a punctured tire.

(7) The present disclosure is a puncture repair liquid unit including a container containing a puncture repair liquid in a space inside the container, and an extraction cap fixed to a mouth of the container, wherein the extraction cap is provided with a first flow path, a second flow path, and a third flow path, the first flow path extends between a first inlet connectable to an external compressor device for generating compressed air and a first outlet opening in the container, the second flow path extends between a second inlet opening in the container and a second outlet for taking out the puncture repair liquid from the extraction cap to supply to a punctured tire to be repaired, the second inlet communicates with the first outlet via the space of the container, and the third flow path communicates with the first flow path and the second flow path outside the space of the container so that a portion of the compressed air in the first flow path flows into the second flow path.

(8) It is possible that the puncture repair liquid unit of the present disclosure shortens the time required to supply compressed air to a punctured tire by adopting the above configuration.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1** is a perspective view illustrating a puncture repair kit of the present embodiment in use.

(2) FIG. **2** is a partial cross-sectional view of a puncture repair unit of the present embodiment.

(3) FIG. **3** is an exploded and enlarged view of FIG. **2**.

(4) FIG. **4** is a partial cross-sectional view showing the puncture repair liquid unit in which an internal cap is moved into a space inside a container.

(5) FIG. **5** is a partial cross-sectional view of the puncture repair liquid unit according to another embodiment of the present disclosure.

(6) FIG. **6** is a cross-sectional view of the puncture repair liquid unit after a closure is moved.

DETAILED DESCRIPTION OF THE INVENTION

(7) An embodiment of the present disclosure will now be described in conjunction with accompanying drawings. It must be understood that the drawings may contain exaggerations or representations different from the actual dimensional ratios of the structure to aid in understanding the contents of the disclosure. Further, throughout the embodiments, identical or common elements are designated by the same reference signs, and redundant explanations are omitted. Furthermore, the embodiments and the specific configurations represented in the drawings are for the purpose of understanding the contents of the present disclosure, and the present disclosure is not limited to the specific configurations shown in the drawings.

(8) (Puncture Repair Kit)

(9) FIG. **1** is a perspective view showing a usage state of a puncture repair kit **1** of the present embodiment. The puncture repair kit **1** of the present embodiment includes a puncture repair liquid unit **2** and a compressor device **3**.

(10) In the puncture repair kit **1** of the present embodiment, the puncture repair liquid unit **2** and the compressor device **3** are connected at the time of puncture repair, for example. Further, in the present embodiment, one end of a hose **4** is connected to the puncture repair liquid unit **2**, for

example. The other end of the hose **4** is connected to a valve **6** of a punctured tire (i.e., flat tire) **5** to be repaired.
(11) (Compressor Device)
(12) The compressor device **3** of the present embodiment is for generating compressed air, and a known one can be employed, for example. In the compressor device **3** of the present embodiment, compressed air (e.g., about 300 to 400 kPa) is generated by the operation of a built-in motor (not shown), for example. The generated compressed air is discharged from a compressed air outlet **7** connectable to the puncture repair liquid unit **2**.
(13) (Puncture Repair Unit (First Embodiment))
(14) The puncture repair liquid unit **2** of the present embodiment has a container **11** and an extraction cap **12**. FIG. **2** shows a cross-sectional view of the container **11** and the extraction cap **12**. In the present embodiment, the configuration of the puncture repair liquid unit **2** is described in a reference posture in which a mouth **16** of the container **11** faces downward.
(15) (Container)
(16) The container **11** of the present embodiment contains puncture repair liquid **14** in a space **13** inside the container **11**. The container **11** of the present embodiment is configured to include a body portion **15** and the mouth **16** as in the above-mentioned Patent Document 1.
(17) The body portion **15** of the present embodiment is formed in a cylindrical shape. The mouth **16** of the present embodiment protrudes from the end portion (lower end) of the body portion **15**. The outer peripheral surface of the mouth **16** has an external thread **17**.
(18) (Extraction Cap)
(19) The extraction cap **12** of the present embodiment includes a base portion **19**, a recessed portion **20**, a boss portion **21**, and a rib portion **22**.
(20) (Base Portion and Recessed Portion)
(21) As shown in FIG. **1**, the base portion **19** of the present embodiment is formed in a cylindrical shape centered around an axial center (not shown) extending in the vertical direction, for example. As shown in FIGS. **1** and **2**, the recessed portion **20** of the present embodiment is recessed from one end (upper end) (**19***t*) of the base portion **19**. The recessed portion **20** of the present embodiment has an inner peripheral surface (**20***a*) and a bottom surface (**20***b*), and is formed in a concave columnar shape centered on the above-mentioned axial center (not shown), for example.
(22) The inner peripheral surface (**20***a*) of the recessed portion **20** in the present embodiment is provided with an internal thread **23** into which the external thread **17** of the container **11** can be screwed. By screwing the external thread **17** into this internal thread **23**, the extraction cap **12** is fixed to the mouth **16** of the container **11**. The bottom surface (**20***b*) of the recessed portion **20** may be provided with a packing material **24** for sealing the end of the mouth **16**.
(23) (Boss Portion)
(24) The boss portion **21** of the present embodiment protrudes from the bottom surface (**20***b*) of the recessed portion **20** toward the container **11** (upward in FIG. **2**) in the container **11**. One end (upper end, protruding end) (**21***t*) of the boss portion **21** in the present embodiment is arranged on the container **11** side (upper side) of the one end (**19***t*) of the base portion **19**.
(25) The boss portion **21** of the present embodiment is arranged substantially in the center of the recessed portion **20**, and is formed in a cylindrical shape centered on an axial center (not shown) extending in the vertical direction, for example. The boss portion **21** of the present embodiment is formed concentrically with the recessed portion **20**, for example.
(26) FIG. **3** is an exploded enlarged view of FIG. **2**. The boss portion **21** of the present embodiment has an outer peripheral surface (**21***a*) provided with a concave groove **25** recessed inward in the radial direction of the boss portion **21**. The concave groove **25** of the present embodiment is formed continuously in the circumferential direction of the boss portion **21**, for example.
(27) (Rib Portion)
(28) As shown in FIG. **2**, the rib portion **22** of the present embodiment protrudes from the bottom

surface (**20***b*) of the recessed portion **20** toward the container **11** (upward in FIG. **2**) in the container **11**. The rib portion **22** of the present embodiment is arranged so as to surround the outer peripheral surface (**21***a*) (shown in FIG. **3**) of the boss portion **21** and is formed in a ring shape (cylindrical shape) centered on an axial center (not shown) extending in the vertical direction, for example. The rib portion **22** of the present embodiment is formed concentrically with the recessed portion **20** and the boss portion **21**, for example. The boss portion **21** and the rib portion **22** may be collectively referred to as a first cylindrical portion.

(29) The extraction cap **12** of the present embodiment includes a first flow path **31**, a second flow path **32**, and a third flow path **33**.

(30) (First Flow Path)

(31) The first flow path **31** of the present embodiment extends between a first inlet **34** and a first outlet **35**. The first inlet **34** is connectable to the compressed air outlet **7** of the compressor device **3** that generates compressed air (C). The first outlet **35** is open in the container **11**.

(32) The first inlet **34** in the present embodiment is formed on a first nozzle portion **39** protruding from an outer peripheral surface (**19***a*) of the base portion **19**, for example. This first inlet **34** (first nozzle portion **39**) is connected to the compressed air outlet **7** (shown in FIG. **1**) of the compressor device **3**.

(33) The first outlet **35** of the present embodiment is formed as a hole opened at the one end (upper end) (**21***t*) of the boss portion **21**, for example. With the first inlet **34** and the first outlet **35** configured as such, the first flow path **31** of the present embodiment can supply the compressed air (C) generated by the compressor device **3** to the container **11** by allowing the compressed air (C) to flow from the first inlet **34** to the first outlet **35**.

(34) The first flow path **31** of the present embodiment includes a first portion **31**A and a second portion **31**B. The first portion **31**A and the second portion **31**B intersect with each other via a first connecting portion **31**C.

(35) The first portion **31**A of the present embodiment extends substantially horizontally between the first inlet **34** and the first connecting portion **31**C, for example. The second portion **31**B of the present embodiment extends substantially vertically between the first connecting portion **31**C and the first outlet **35**, for example. By the first portion **31**A and the second portion **31**B configured as such, the first flow path **31** of the present embodiment is formed in an L-shape. Thereby, the first flow path **31** can change (bend) the flow direction of the compressed air (C) supplied from the first inlet **34** at the first connecting portion **31**C to let the compressed air (C) flow to the first outlet **35**, for example. The first portion **31**A and the second portion **31**B of the present embodiment are formed to have substantially the same inner diameter R1 (shown in FIG. **3**) except for the first inlet **34**, for example.

(36) (Second Flow Path)

(37) The second flow path **32** of the present embodiment extends between a second inlet **36** opened in the container **11** and a second outlet **37** for taking out the puncture repair liquid **14** from the extraction cap **12** to supply to the punctured tire **5** (shown in FIG. **1**) to be repaired.

(38) The second inlet **36** of the present embodiment is formed as an annular hole provided between the boss portion **21** and the rib portion **22**. This second inlet **36** communicates with the first outlet **35** via the space **13** of the container **11** when an internal cap **41** described later is moved to the space **13** of the container **11** (that is, the internal cap **41** is removed), for example.

(39) The second outlet **37** of the present embodiment is formed in a second nozzle portion **40** protruding from the outer peripheral surface (**19***a*) of the base portion **19**, for example. The second outlet **37** (second nozzle portion **40**) is to be connected to the hose **4** (shown in FIG. **1**) connected to the punctured tire **5** to be repaired, for example.

(40) With the second inlet **36** and the second outlet **37** configured as such, the second flow path **32** of the present embodiment allows the puncture repair liquid **14** contained in the container **11** to flow from the second inlet **36** to the second outlet **37**. As a result, the puncture repair liquid **14** can

be supplied to the punctured tire **5** (shown in FIG. **1**) to be repaired.
(41) The second flow path **32** of the present embodiment includes a third portion **32**A and a fourth portion **32**B. The third portion **32**A and the fourth portion **32**B intersect with each other via a second connecting portion **32**C.
(42) The third portion **32**A of the present embodiment extends substantially vertically, for example, between the second inlet **36** and the second connecting portion **32**C. The fourth portion **32**B of the present embodiment extends substantially horizontally, for example, between the second connecting portion **32**C and the second outlet **37**. By the third portion **32**A and the fourth portion **32**B configured as such, the second flow path **32** of the present embodiment is formed in an L-shape. Thereby, the second flow path **32** can change (bend) the flow direction of the puncture repair liquid **14** supplied from the second inlet **36** at the second connecting portion **32**C to let the puncture repair liquid **14** flow to the second outlet **37**, for example. The fourth portion **32**B of the present embodiment is configured to have an inner diameter larger than an inner diameter of the third portion **32**A, for example.
(43) (Third Flow Path)
(44) The third flow path **33** of the present embodiment communicates the first flow path **31** and the second flow path **32** outside the space **13** of the container **11** (inside the extraction cap **12** in the present example). The third flow path **33** configured as such allows a part of the compressed air (C) in the first flow path **31** to flow to the second flow path **32** without going through the space **13** of the container **11**.
(45) The third flow path **33** of the present embodiment extends substantially horizontally between the first connecting portion **31**C of the first flow path **31** and the second connecting portion **32**C of the second flow path **32**, for example. Therefore, in the present embodiment, the first portion **31**A of the first flow path **31**, the fourth portion **32**B of the second flow path **32**, and the third flow path **33** are arranged to form a straight line. As a result, the third flow path **33** can allow a part of the compressed air (C) in the first flow path **31** to smoothly flow to the second flow path **32**.
(46) (Internal Cap)
(47) The first outlet **35** and the second inlet **36** of the present embodiment are separated from the space **13** of the container **11** by the internal cap **41**. The internal cap **41** of the present embodiment is detachably attached to the boss portion **21** and the rib portion **22** of the extraction cap **12**, for example.
(48) As shown in FIG. **3**, the internal cap **41** of the present embodiment includes a first portion **41**A, a second portion **41**B, a third portion **41**C, and a fourth portion **41**D. The internal cap **41** of the present embodiment is formed in a cylindrical shape capable of accommodating the boss portion **21** and the rib portion **22**.
(49) The first portion **41**A of the present embodiment is for covering the first outlet **35** (the one end (upper end) (**21***t*) of the boss portion **21**). The first portion **41**A is formed in a conical shape, for example. The second portion **41**B of the present embodiment is for covering the one end (**21***t*) side portion of the outer peripheral surface (**21***a*) of the boss portion **21**, for example. The second portion **41**B is formed in a cylindrical shape, for example.
(50) The third portion **41**C of the present embodiment is for covering the other end side (bottom surface (**20***b*) side of the recessed portion **20**) portion of the outer peripheral surface (**21***a*) of the boss portion **21** and the rib portion **22**. The third portion **41**C is formed in a cylindrical shape, for example. The third portion **41**C of the present embodiment is formed to have an inner diameter larger than an inner diameter of the second portion **41**B.
(51) The fourth portion **41**D in the present embodiment extends between the second portion **41**B and the third portion **41**C, for example. The fourth portion **41**D of the present embodiment has an inner diameter gradually and continuously decreasing from the third portion **41**C to the second portion **41**B, and thus is formed in a tapered shape, for example.
(52) The internal cap **41** of the present embodiment is provided with a ridge portion **42** protruding

inward in a radial direction of the internal cap **41** on an inner peripheral surface (**41***a*) of the internal cap **41**. The ridge portion **42** of the present embodiment includes a first ridge portion **42**A and a second ridge portion (another ridge portion) **42**B. The first ridge portion **42**A and the second ridge portion **42**B are continuously formed in a circumferential direction of the internal cap **41**.
(53) The first ridge portion **42**A of the present embodiment protrudes toward the boss portion **21** (radially inward of the internal cap **41**). The first ridge portion **42**A in the present embodiment is fitted into the concave groove **25** of the boss portion **21**. This allows the first ridge portion **42**A to secure the internal cap **41** to the boss portion **21** while sealing between the internal cap **41** and the boss portion **21**. It should be noted that the boss portion **21** may be provided with a ridge portion protruding radially outward and the internal cap **41** may be provided with a concave groove recessed radially outward.
(54) The second ridge portion **42**B of the present embodiment protrudes toward the rib portion **22** (radially inward of the internal cap **41**). The second ridge portion **42**B of the present embodiment is in contact with an outer peripheral surface (**22***a*) of the rib portion **22** over the entire circumference thereof. This allows the second ridge portion **42**B to secure the internal cap **41** to the rib portion **22** while sealing between the internal cap **41** and the rib portion **22**. It should be noted that the second ridge portion may be provided on the outer peripheral surface of the rib portion **22** to be in contact with the inner peripheral surface of the internal cap **41**. It should be noted that the second ridge portion may be provided on the outer peripheral surface (**22***a*) of the rib portion **22** instead to be in contact with the inner peripheral surface (**41***a*) of the internal cap **41**, specifically the inner peripheral surface of the third portion **41**C.
(55) In this way, the internal cap **41** of the present embodiment is fixed to the boss portion **21** and the rib portion **22**, therefore, the first outlet **35** and the second inlet **36** can be separated from the space **13** of the container **11**. As a result, the internal cap **41** can prevent the puncture repair liquid **14** from leaking out of the container **11** during storage before puncture repair, for example. The first portion **41**A, the second portion **41**B, the third portion **41**C, and the fourth portion **41**D (i.e., the internal cap **41** in the present embodiment) may be collectively referred to as a second cylindrical portion.
(56) (Workings of Puncture Repair Liquid Unit and Puncture Repair Kit)
(57) Next, workings of the puncture repair liquid unit **2** and the puncture repair kit **1** of the present embodiment will be described. In the present embodiment, first, as shown in FIG. **1**, the puncture repair liquid unit **2** and the compressor device **3** are connected, and the valve **6** of the punctured tire **5** and the hose **4** are connected. Next, the compressor device **3** starts generating the compressed air (C) (shown in FIG. **2**). This allows the puncture repair liquid unit **2** (the puncture repair kit **1**) of the present embodiment to supply the compressed air (C) to the first inlet **34** and to flow the compressed air (C) into the first flow path **31** (the first portion **31**A) as shown in FIG. **2**.
(58) The compressed air (C) flowing in the first flow path **31** (the first portion **31**A) splits into the second portion **31**B of the first flow path **31** and the third flow path **33** at the first connecting portion **31**C. The compressed air (C) flowing in the third flow path **33** flows in the second flow path **32** (the fourth portion **32**B) without going through the space **13** of the container **11**.
(59) On the other hand, the compressed air (C) flowing into the second portion **31**B of the first flow path **31** flows out of the first outlet **35** into a space **43** enclosed by the internal cap **41**, the boss portion **21**, and the rib portion **22**, causing the internal cap **41** to expand. Due to the expansion of the internal cap **41**, in the present embodiment, the fixation between the internal cap **41** and the boss portion **21** and the rib portion **22** can be released, and the internal cap **41** can be moved away from the first outlet **35** and the second inlet **36** into the space **13** of the container **11**.
(60) FIG. **4** is a cross-sectional view showing the puncture repair liquid unit **2** in which the internal cap **41** is moved into the space **13**. In the present embodiment, the compressed air (C) can be supplied to the space **13** inside the container **11** by the movement of the internal cap **41**. Further, it is possible that the first outlet **35** and the second inlet **36** communicate with the space **13** in the

present embodiment.
(61) In the puncture repair liquid unit **2** (the puncture repair kit **1**) of the present embodiment, the puncture repair liquid **14** contained in the container **11** can flow from the second inlet **36** to the second outlet **37** (shown in FIG. **2**) by supplying the compressed air (C) to the space **13** of the container **11**. Thereby, in the present embodiment, the puncture repair liquid **14** can be taken out from the extraction cap **12** and supplied to the punctured tire **5** (shown in FIG. **1**) to be repaired.
(62) Further, in the present embodiment, since a part of the compressed air (C) in the first flow path **31** flows to the second flow path **32** via the third flow path **33**, the pressure of the compressed air (C) supplied to the container **11** from the first flow path **31** (the second portion **31**B) can be reduced compared to the conventional unit. As a result, the puncture repair liquid unit **2** (the puncture repair kit **1**) of the present embodiment can gradually supply the puncture repair liquid **14** to the punctured tire **5** unlike conventional units in which the puncture repair liquid **14** is supplied to the punctured tire **5** (shown in FIG. **1**) all at once.
(63) As a result of diligent research by the disclosers, it was found that in the conventional unit in which the puncture repair liquid **14** is supplied at once, a part of the puncture repair liquid **14** tends to solidify near the valve **6** (shown in FIG. **1**) of the punctured tire **5**, which subsequently causes blocking of a part of the compressed air (C) supply path. This tendency is more pronounced in a high temperature environment (50 to 70° C., for example, including operation and storage environments). Due to such blockage of the supply path, there has been a problem that it takes a lot of time to supply the compressed air (C) to the punctured tire **5**.
(64) On the other hand, the puncture repair liquid unit **2** (the puncture repair kit **1**) of the present embodiment can gradually supply the puncture repair liquid **14** to the punctured tire **5** (shown in FIG. **1**), and therefore it is possible that the puncture repair liquid **14** is prevented from hardening near the valve **6** (shown in FIG. **1**). As a result, in the present embodiment, the compressed air (C) can be stably supplied to the punctured tire **5**.
(65) Further, in the present embodiment, since a part of the compressed air (C) in the first flow path **31** can be flowed to the second flow path **32** via the third flow path **33**, it is possible that the compressed air (C) is supplied together with the puncture repair liquid **14** to the punctured tire **5**. As a result, the puncture repair liquid unit **2** (the puncture repair kit **1**) of the present embodiment can shorten the supply time of the compressed air (C) to the punctured tire **5**.
(66) The first flow path **31** of the present embodiment allows the compressed air (C) supplied from the first inlet **34** (shown in FIG. **2**) to change (bend) the flow direction of the compressed air (C) at the first connecting portion **31**C to flow to the first outlet **35**. Further, the third flow path **33** of the present embodiment allows a part of the compressed air (C) in the first flow path **31** to linearly flow to the second flow path **32**. As a result, in the present embodiment, the pressure of the compressed air (C) supplied from the first flow path **31** to the container **11** can be effectively prevented from becoming higher than necessary, therefore, it is possible that the puncture repair liquid **14** is more reliably prevented from being supplied all at once.
(67) As shown in FIG. **3**, it is preferred that the third flow path **33** is formed to have an inner diameter R3 smaller than the inner diameter R1 of the first flow path **31**. Thereby, the amount of the compressed air (C) flowing from the third flow path **33** to the second flow path **32** can be prevented from become excessively large, thereby, it is possible that the pressure of the compressed air (C) in the first flow path **31** (the second portion **31**B) is prevented from decreasing more than necessary. As a result, in the present embodiment, for example, during puncture repair in a low temperature environment (for example, −30° C. or lower), the internal cap **41**, which tends to be strongly fixed, can be easily moved, therefore, it is possible that the supply time of the compressed air (C) is shortened. The inner diameter R3 is the maximum diameter of the third flow path **33** and the inner diameter R1 is the maximum diameter of the first flow path **31** (excluding the first inlet **34**) in the present embodiment.
(68) It is preferred that a ratio (R1/R3) of the inner diameters is set in the range of 1.25 or more and

3.50 or less. By setting the ratio (R1/R3) to 1.25 or higher, the pressure of the compressed air (C) supplied to the space **13** inside the container **11** can be prevented from decreasing more than necessary, thereby, the internal cap **41** can be easily moved when repairing a flat tire in a low temperature environment. On the other hand, by setting the ratio (R1/R3) to 3.50 or lower, the pressure of the compressed air (C) supplied to the space **13** inside the container **11** can be prevented from becoming larger than necessary, thereby, it is possible that the puncture repair liquid **14** is prevented from solidifying near the valve **6**. From this point of view, the ratio (R1/R3) is preferably 1.75 or more, and preferably 2.80 or less.
(69) (Puncture Repair Liquid Unit (Second Embodiment))
(70) FIG. **5** is a cross-sectional view of the puncture repair liquid unit **2** according to another embodiment of the present disclosure. The same configurations as those of the previous embodiment are indicated by the same reference signs, and the description thereof may be omitted.
(71) The first outlet **35** of this embodiment is formed as a hole opened in the outer peripheral surface (**21***a*) of the boss portion **21**. The first outlet **35** of this embodiment is inclined toward the bottom surface (**20***b*) of the recessed portion **20** as it goes from the first flow path **31** (the second portion **31**B in this embodiment) to the outer peripheral surface (**21***a*). Further, the first outlet **35** is provided between the one end (**21***t*) and the concave groove **25** of the boss portion **21**. In this embodiment, one first outlet **35** is provided, but a plurality of the first outlets (not shown) may be provided.
(72) The first flow path **31** of this embodiment is provided with a closure **45** that closes (blocks) the first flow path **31** between the first outlet **35** and the first inlet **34** (shown in FIG. **2**). The closure **45** is formed in a cylindrical shape centered on an axial center (not shown) extending in the vertical direction, and is press-fitted into the first flow path **31** (the second portion **31**B in this embodiment).
(73) Since the closure **45** of this embodiment is arranged between the first outlet **35** and the first inlet **34** (shown in FIG. **2**), the closure **45** can close the path between the first outlet **35** and the first inlet **34** during storage before puncture repair, for example. Thereby, the closure **45** can prevent the puncture repair liquid **14** from flowing back into the first flow path **31** even if the internal cap **41** comes off due to mishandling, for example.
(74) The closure **45** of this embodiment is moved in a first direction D**1** (the longitudinal direction of the second portion **31**B (vertical direction) in this embodiment) due to the compressed air (C) flowing through the second portion **31**B by the supply of the compressed air (C) to the first inlet **34** at the time of puncture repair. This movement of the closure **45** allows the first inlet **34** (shown in FIG. **2**) to communicate with the first outlet **35**. In order to ensure smooth movement of the closure **45** in this way, it is preferred that the ratio (R1/R3) of the inner diameters is set in the range of 1.25 or more and 3.50 or less. FIG. **6** is a cross-sectional view of the puncture repair liquid unit **2** after the closure **45** is moved.
(75) Due to the communication between the first inlet **34** (shown in FIG. **2**) and the first outlet **35**, the compressed air (C) flowing in the second portion **31**B flows from the first outlet **35** to the space **43** surrounded by the internal cap **41** and the boss portion **21** and the rib portion **22**. Thereby, in this embodiment, it is possible that the internal cap **41** is expanded and moved into the space **13** of the container **11** in the same manner as in the previous embodiment shown in FIG. **4**.
(76) The puncture repair liquid unit **2** (the puncture repair kit **1**) of this embodiment can supply the compressed air (C) to the space **13** inside the container **11** as in the previous embodiment. Further, it is possible that the first outlet **35** and the second inlet **36** communicate with the space **13**. Thereby, in this embodiment, as in the previous embodiment, the puncture repair liquid **14** can be taken out from the extraction cap **12** to supply to the punctured tire **5** (shown in FIG. **1**) to be repaired, therefore, it is possible that the supply time of the compressed air (C) to the punctured tire **5** is shortened.
(77) As shown in FIG. **5**, it is preferred that an inner diameter (R1a) of the first outlet **35** is set

smaller than the inner diameter R1 of the first flow path **31**. This allows the pressure of the compressed air (C) supplied from the first outlet **35** to be increased in this embodiment, therefore, the internal cap **41** is reliably moved to the space **13** of the container **11** and it is possible that the puncture repair liquid **14** is gradually supplied to the punctured tire **5**.
(78) It is preferred that a ratio (R1a/R1) of the inner diameters is set in the range of 0.2 or more and 0.7 or less. By setting the ratio R1a/R1 to 0.2 or more, it is possible that the puncture repair liquid **14** is gradually supplied to the punctured tire **5** while the internal cap **41** is reliably moved into the space **13** of the container **11**. On the other hand, by setting the ratio (R1a/R1) to 0.7 or less, it is possible that the puncture repair liquid **14** is prevented from being supplied to the punctured tire **5** all at once. From such a point of view, the ratio (R1a/R1) is preferably 0.3 or more and preferably 0.6 or less.
(79) As shown in FIGS. **5** and **6**, it is preferred that the first flow path **31** is provided with a retaining portion **46** to prevent the closure **45** from moving to the space **13** of the container **11**. Thereby, this prevents the puncture repair liquid unit **2** from having an outlet (not shown) larger than the inner diameter (R1a) in the first flow path **31**, apart from the first outlet **35**. Therefore, in this embodiment, it is possible that the puncture repair liquid **14** is prevented from being supplied to the punctured tire **5** (shown in FIG. **1**) at once.
(80) While detailed description has been made of the especially preferred embodiments of the present disclosure, the present disclosure can be embodied in various forms without being limited to the illustrated embodiments.
EXAMPLES
Example A
(81) The puncture repair liquid units and the puncture repair kits having the basic structure shown in FIG. **1** and FIG. **2** and each having the first flow path, the second flow path, and the third flow path were made by way of test (prototyped) (Examples 1 to 6). For comparison, the puncture repair liquid unit and the puncture repair kit not having the third flow path were made by way of test (Reference).
(82) Next, compressed air (350 kPa) was generated by the compressor device and the punctured tires were repaired by using the prototyped puncture repair kits. Then, the presence or absence of the solidification of the puncture repair liquid near the valve, the moving time of the inner cap in a low temperature environment, and the supply time of the compressed air were evaluated. The test methods were as follows. The test results are shown in Table 1.
(83) (Presence or Absence of Solidification of Puncture Repair Liquid)
(84) After storing the prototype puncture repair kits in a high temperature environment (60° C.) for 24 hours, the puncture repair liquid was supplied to the puncture tires. After all the puncture repair liquid was supplied, it was visually confirmed whether or not the puncture repair liquid was solidified near the valve. In the results, one without the solidification of the puncture repair liquid is indicated as “No” and ones with the solidification of the puncture repair liquid are indicated as “Yes”.
(85) (Moving Time of Inner Cap)
(86) After storing the prototyped puncture repair kits in a low temperature environment (−40° C.) for 24 hours, compressed air was generated by the compressor device, and the time (duration) until the inner cap was moved into the space of the container was measured for each prototyped kit. The results are graded as follows. Excellent: moved within 1 minute Good: moved in 1 to 3 minutes Fair: moved in 3 to 5 minutes
(Supply Time of Compressed Air)
(87) After the prototype puncture repair kits were stored in a high temperature environment (60° C.) for 24 hours, the time (duration) from the beginning of the supply of the puncture repair liquid to the punctured tire until the punctured tire was inflated to the predetermined internal pressure was measured for each prototype kit. The results are indicated by an index based on Reference being

100, wherein the smaller numerical value is better.

(88) TABLE-US-00001 TABLE 1 Ref. Ex. 1 Ex. 2 Ex. 3 Ex. 4 Ex. 5 Ex. 6 Presence or Absence of Third flow path Absence Presence Presence Presence Presence Presence Presence Inner diameter R1 of First flow path [mm] 3.5 3.5 3.5 3.5 3.5 3.5 3.5 Inner diameter R3 of Third flow path [mm] — 0.5 1.0 2.0 2.8 3.5 4.0 Ratio (R1/R3) — 7.00 3.50 1.75 1.25 1.00 0.88 Presence or Absence Presence Absence Absence Absence Absence Absence Absence of Solidification of Puncture repair liquid Moving time of Internal cap Excellent Excellent Excellent Excellent Excellent Good Fair Supply time of Compressed air [index] 100 90 80 75 79 82 87

(89) From the test results, it was confirmed that the puncture repair liquid can be prevented from solidifying near the valve and thus the supply time of the compressed air to the punctured tires can be shortened in the Examples compared to the Reference. Further, it was confirmed that the moving time of the inner cap under the low temperature environment can be shortened in Examples 2 to 4 in which the ratio (R1/R3) of the inner diameter R3 of the third flow path and the inner diameter R1 of the first flow path was within the preferred range compared to Examples 1, 5, and 6 in which the ratio (R1/R3) was out of the preferred range.

Example B

(90) The puncture repair liquid units and the puncture repair kits having the basic structure shown in FIG. **1** and FIG. **5** and each having the first flow path, the second flow path, the third flow path, and the closure were made by way of test (prototyped) (Examples 7 to 12). Next, compressed air (350 kPa) was generated by the compressor device and the punctured tires were repaired by using the prototyped puncture repair kits. Then, the presence or absence of the solidification of the puncture repair liquid near the valve, and the moving time of the closure in a low temperature environment were evaluated. The test methods were as follows except for the presence or absence of the solidification of the puncture repair liquid near the valve. The test results are shown in Table 2.

(91) (Moving Time of Closure)

(92) After the prototype puncture repair kits were stored in a low temperature environment (−40° C.) for 24 hours, the moving time of the closure was measured from the beginning of the generation of the compressed air by the compressor device until the first inlet communicates with the first outlet. The results are graded as follows. Excellent: moved within 1 minute Good: moved in 1 to 3 minutes

(93) TABLE-US-00002 TABLE 2 Ex. 7 Ex. 8 Ex. 9 Ex. 10 Ex. 11 Ex. 12 Inner diameter R1 of First flow path [mm] 3.5 3.5 3.5 3.5 3.5 3.5 Inner diameter R3 of Third flow path [mm] 0.5 1.0 2.0 2.8 3.5 4.0 Ratio (R1/R3) 7.00 3.50 1.75 1.25 1.00 0.88 Presence or Absence Absence Absence Absence Absence Absence Absence of Solidification of Puncture repair liquid Moving time of Closure Excellent Excellent Excellent Excellent Good Good Supply time of Compressed air [index] 90 80 75 79 82 87

(94) From the test results, it was confirmed that the puncture repair liquid can be prevented from solidifying near the valve and thus the supply time of the compressed air to the punctured tires can be shortened in Examples 7 to 12 similarly to Examples 1 to 6 of Table 1. Further, it was confirmed that the moving time of the closure under the low temperature environment can be shortened in Examples 8 to 10 in which the ratio (R1/R3) of the inner diameter R3 of the third flow path and the inner diameter R1 of the first flow path was within the preferred range compared to Examples 11 and 12 in which the ratio (R1/R3) was out of the preferred range.

Statement of Disclosure

(95) The present disclosure includes the following aspects.

Present Disclosure 1

(96) A puncture repair liquid unit including: a container containing a puncture repair liquid in a space inside the container; and an extraction cap fixed to a mouth of the container, wherein the extraction cap is provided with a first flow path, a second flow path, and a third flow path, the first

flow path extends between a first inlet connectable to an external compressor device for generating compressed air and a first outlet opening in the container, the second flow path extends between a second inlet opening in the container and a second outlet for taking out the puncture repair liquid from the extraction cap to supply to a punctured tire to be repaired, the second inlet communicates with the first outlet via the space of the container, and the third flow path communicates with the first flow path and the second flow path outside the space of the container so that a portion of the compressed air in the first flow path flows into the second flow path.

Present Disclosure 2

(97) The puncture repair liquid unit according to present disclosure 1, wherein the third flow path has an inner diameter (R3) smaller than an inner diameter (R1) of the first flow path.

Present Disclosure 3

(98) The puncture repair liquid unit according to present disclosure 2, wherein a ratio (R1/R3) of the inner diameter (R1) of the first flow path to the inner diameter (R3) of the third flow path is 1.25 or higher and 3.50 or lower.

Present Disclosure 4

(99) The puncture repair liquid unit according to any one of present disclosures 1 to 3 further comprising an internal cap for separating the first outlet and the second inlet from the space of the container, wherein the first outlet and the second inlet communicate with the space when the internal cap is moved into the space of the container by the compressed air supplied to the first inlet.

Present Disclosure 5

(100) The puncture repair liquid unit according to present disclosure 4, wherein the first flow path is provided with a closure for closing the first flow path between the first outlet and the first inlet, and the first inlet and the first outlet communicate with each other when the closure is moved in a first direction toward the first outlet to pass the first outlet by the compressed air supplied to the first inlet.

Present Disclosure 6

(101) A puncture repair kit including the puncture repair liquid unit according to any one of present disclosures 1 to 5 and 7 to 20 and a compressor device.

Present Disclosure 7

(102) The puncture repair liquid unit according to claim **2**, wherein the ratio (R1/R3) of the inner diameter (R1) of the first flow path to the inner diameter (R3) of the third flow path is 1.75 or higher and 2.80 or lower.

Present Disclosure 8

(103) The puncture repair liquid unit according to present disclosure 4, wherein the extraction cap includes a first cylindrical portion protruding in a direction from the mouth to further inside of the container, the extraction cap includes a second cylindrical portion for covering the first cylindrical portion, the first inlet and the second inlet are provided in the first cylindrical portion, the first cylindrical portion has an outer peripheral surface provided with one of a concave groove and a ridge portion extending continuously in a circumferential direction of the first cylindrical portion in a ring shape, the second cylindrical portion has an inner peripheral surface provided with the other of the concave groove and the ridge portion extending continuously in a circumferential direction of the second cylindrical portion in a ring shape, and the one and the other of the concave groove and the ridge portion are detachably fitted with each other before the compressed air is supplied to the first inlet.

Present Disclosure 9

(104) The puncture repair liquid unit according to present disclosure 8, wherein the one of the concave groove and the ridge portion is provided in the first cylindrical portion between the first outlet and the second inlet, and the first outlet opens in a space surrounded by the first cylindrical portion and the second cylindrical portion with the one and the other of the concave groove and the

ridge portion being fitted with each other.
Present Disclosure 10
(105) The puncture repair liquid unit according to present disclosure 9, wherein one of the outer peripheral surface of the first cylindrical portion and the inner peripheral surface of the second cylindrical portion is provided with another ridge portion extending continuously in the circumferential direction in a ring shape, said another ridge portion is in contact with the other of the outer peripheral surface of the first cylindrical portion and the inner peripheral portion of the second cylindrical portion over the entire circumference thereof, and the second inlet is arranged between the one of the concave groove and the ridge portion and said another ridge portion.
Present Disclosure 11
(106) The puncture repair liquid unit according to present disclosure 1, wherein the first flow path has a first portion and a second portion, the first portion includes the first inlet and extends linearly, the second portion includes the first outlet and extends linearly, and the first portion and the second portion are connected via a first connecting portion in an L-shape.
Present Disclosure 12
(107) The puncture repair liquid unit according to present disclosure 11, wherein the second flow path has a third portion and a fourth portion, the third portion includes the second inlet and extends linearly, the fourth portion includes the second outlet and extends linearly, and the third portion and the fourth portion are connected via a second connecting portion in an L-shape.
Present Disclosure 13
(108) The puncture repair liquid unit according to present disclosure 12, wherein the third flow path communicate with the first portion and the fourth portion, and the first portion, the third flow path, and the fourth portion are arranged to form a single liner line.
Present Disclosure 14
(109) The puncture repair liquid unit according to present disclosure 9, wherein the first outlet is formed at a protruding end of the first cylindrical portion.
Present Disclosure 15
(110) The puncture repair liquid unit according to present disclosure 9, wherein the first flow path is provided with a closure for closing the first flow path between the first outlet and the first inlet, the first inlet and the first outlet communicate with each other when the closure is moved in a first direction toward the first outlet to pass the first outlet by the compressed air supplied to the first inlet, and the first outlet is formed as a through hole penetrating between an inner peripheral surface and the outer peripheral surface of the first cylindrical portion and arranged between a protruding end of the first cylindrical portion and the one of the concave groove and the ridge portion.
Present Disclosure 16
(111) The puncture repair liquid unit according to present disclosure 15, wherein the first outlet is inclined in a direction away from the protruding end as it goes radially outward.
Present Disclosure 17
(112) The puncture repair liquid unit according to present disclosure 15, wherein the first cylindrical portion is provided with a plurality of the first outlets.
Present Disclosure 18
(113) The puncture repair liquid unit according to present disclosure 15, wherein the first outlet has an inner diameter (Ria) smaller than an inner diameter (R1) of the first flow path.
Present Disclosure 19
(114) The puncture repair liquid unit according to present disclosure 18, wherein a ratio (R1a/R1) of the inner diameter (R1a) of the first outlet to the inner diameter (R1) of the first flow path is 0.2 or higher and 0.7 or lower.
Present Disclosure 20
(115) The puncture repair liquid unit according to present disclosure 15, wherein the first

cylindrical portion is provided with a retaining portion for stopping the movement of the closure, and the retaining portion is arranged on the protruding end side with respect to the first outlet.
DESCRIPTION OF THE REFERENCE SIGNS
(116) **1** puncture repair kit **2** puncture repair liquid unit **3** compressor device **5** punctured tire **11** container **12** extraction cap **14** puncture repair liquid **31** first flow path **32** second flow path **33** third flow path

Claims

1. A puncture repair liquid unit comprising: a container containing a puncture repair liquid in a space inside the container; and an extraction cap fixed to a mouth of the container, wherein the extraction cap is provided with a first flow path, a second flow path, and a third flow path, the first flow path extends between a first inlet connectable to an external compressor device for generating compressed air and a first outlet opening in the container, the second flow path extends between a second inlet opening in the container and a second outlet for taking out the puncture repair liquid from the extraction cap to supply to a punctured tire to be repaired, the second inlet communicates with the first outlet via the space of the container, the third flow path communicates with the first flow path and the second flow path outside the space of the container so that a portion of the compressed air in the first flow path flows into the second flow path, and the first outlet is located more inwardly in the container relative to the mouth than the second inlet.
2. The puncture repair liquid unit according to claim 1 further comprising an internal cap for separating the first outlet and the second inlet from the space of the container, wherein the first outlet and the second inlet communicate with the space when the internal cap is moved into the space of the container by the compressed air supplied to the first inlet.
3. The puncture repair liquid unit according to claim 2, wherein the internal cap consists of a first portion, a second portion, a third portion, and a fourth portion, and the first portion is located innermost of the container among the first, second, third, and fourth portions, and is formed in a conical shape having a diameter continuously decreasing in a direction from the mouth to the inside of the container in a state in which the first outlet and the second inlet are separated from the space of the container by the internal cap.
4. The puncture repair liquid unit according to claim 2, wherein the first flow path is provided with a closure for closing the first flow path between the first outlet and the first inlet, and the first inlet and the first outlet communicate with each other when the closure is moved in a first direction toward the first outlet to pass the first outlet by the compressed air supplied to the first inlet.
5. The puncture repair liquid unit according to claim 2, wherein the extraction cap includes a first cylindrical portion protruding in a direction from the mouth to further inside of the container, the extraction cap includes a second cylindrical portion for covering the first cylindrical portion, the first inlet and the second inlet are provided in the first cylindrical portion, the first cylindrical portion has an outer peripheral surface provided with one of a concave groove and a ridge portion extending continuously in a circumferential direction of the first cylindrical portion in a ring shape, the second cylindrical portion has an inner peripheral surface provided with the other of the concave groove and the ridge portion extending continuously in a circumferential direction of the second cylindrical portion in a ring shape, and the one and the other of the concave groove and the ridge portion are detachably fitted with each other before the compressed air is supplied to the first inlet.
6. The puncture repair liquid unit according to claim 5, wherein the one of the concave groove and the ridge portion is provided in the first cylindrical portion between the first outlet and the second inlet, and the first outlet opens in a space surrounded by the first cylindrical portion and the second cylindrical portion with the one and the other of the concave groove and the ridge portion being fitted with each other.

7. The puncture repair liquid unit according to claim 6, wherein one of the outer peripheral surface of the first cylindrical portion and the inner peripheral surface of the second cylindrical portion is provided with another ridge portion extending continuously in the circumferential direction in a ring shape, said another ridge portion is in contact with the other of the outer peripheral surface of the first cylindrical portion and the inner peripheral portion of the second cylindrical portion over the entire circumference thereof, and the second inlet is arranged between the one of the concave groove and the ridge portion and said another ridge portion.
8. The puncture repair liquid unit according to claim 6, wherein the first outlet is formed at a protruding end of the first cylindrical portion.
9. The puncture repair liquid unit according to claim 6, wherein the first flow path is provided with a closure for closing the first flow path between the first outlet and the first inlet, the first inlet and the first outlet communicate with each other when the closure is moved in a first direction toward the first outlet to pass the first outlet by the compressed air supplied to the first inlet, and the first outlet is formed as a through hole penetrating between an inner peripheral surface and the outer peripheral surface of the first cylindrical portion and arranged between a protruding end of the first cylindrical portion and the one of the concave groove and the ridge portion.
10. The puncture repair liquid unit according to claim 9, wherein the first cylindrical portion is provided with a retaining portion for stopping the movement of the closure, and the retaining portion is arranged on the protruding end side with respect to the first outlet.
11. The puncture repair liquid unit according to claim 9, wherein the first outlet is inclined in a direction away from the protruding end as it goes radially outward.
12. The puncture repair liquid unit according to claim 9, wherein the first cylindrical portion is provided with a plurality of the first outlets.
13. The puncture repair liquid unit according to claim 9, wherein the first outlet has an inner diameter (R1a) smaller than an inner diameter (R1) of the first flow path.
14. The puncture repair liquid unit according to claim 13, wherein a ratio (R1a/R1) of the inner diameter (R1a) of the first outlet to the inner diameter (R1) of the first flow path is 0.2 or higher and 0.7 or lower.
15. A puncture repair kit including the puncture repair liquid unit according to claim 1 and a compressor device.
16. The puncture repair liquid unit according to claim 1, wherein the first flow path has a first portion and a second portion, the first portion includes the first inlet and extends linearly, the second portion includes the first outlet and extends linearly, and the first portion and the second portion are connected via a first connecting portion in an L-shape.
17. The puncture repair liquid unit according to claim 16, wherein the second flow path has a third portion and a fourth portion, the third portion includes the second inlet and extends linearly, the fourth portion includes the second outlet and extends linearly, and the third portion and the fourth portion are connected via a second connecting portion in an L-shape.
18. The puncture repair liquid unit according to claim 17, wherein the third flow path communicate with the first portion and the fourth portion, and the first portion, the third flow path, and the fourth portion are arranged to form a single line.
19. A puncture repair liquid unit comprising: a container containing a puncture repair liquid in a space inside the container; an extraction cap fixed to a mouth of the container; and an internal cap for separating a first outlet opening in the container and a second inlet opening in the container from the space of the container, wherein the extraction cap is provided with a first flow path, a second flow path, and a third flow path, the first flow path extends between a first inlet connectable to an external compressor device for generating compressed air and the first outlet, the second flow path extends between the second inlet and a second outlet for taking out the puncture repair liquid from the extraction cap to supply to a punctured tire to be repaired, the second inlet communicates with the first outlet via the space of the container, the third flow path communicates with the first

flow path and the second flow path outside the space of the container so that a portion of the compressed air in the first flow path flows into the second flow path, the first outlet and the second inlet communicate with the space when the internal cap is moved into the space of the container by the compressed air supplied to the first inlet, the extraction cap includes a first cylindrical portion protruding in a direction from the mouth to further inside of the container, the extraction cap includes a second cylindrical portion for covering the first cylindrical portion, the first inlet and the second inlet are provided in the first cylindrical portion, the first cylindrical portion has an outer peripheral surface provided with one of a concave groove and a ridge portion extending continuously in a circumferential direction of the first cylindrical portion in a ring shape, the second cylindrical portion has an inner peripheral surface provided with the other of the concave groove and the ridge portion extending continuously in a circumferential direction of the second cylindrical portion in a ring shape, the one and the other of the concave groove and the ridge portion are detachably fitted with each other before the compressed air is supplied to the first inlet, the one of the concave groove and the ridge portion is provided in the first cylindrical portion between the first outlet and the second inlet, the first outlet opens in a space surrounded by the first cylindrical portion and the second cylindrical portion with the one and the other of the concave groove and the ridge portion being fitted with each other, the first flow path is provided with a closure for closing the first flow path between the first outlet and the first inlet, the first inlet and the first outlet communicate with each other when the closure is moved in a first direction toward the first outlet to pass the first outlet by the compressed air supplied to the first inlet, and the first outlet is formed as a through hole penetrating between an inner peripheral surface and the outer peripheral surface of the first cylindrical portion and arranged between a protruding end of the first cylindrical portion and the one of the concave groove and the ridge portion.

20. A puncture repair liquid unit comprising: a container containing a puncture repair liquid in a space inside the container; an extraction cap fixed to a mouth of the container; and an internal cap for separating a first outlet opening in the container and a second inlet opening in the container from the space inside the container, wherein the extraction cap is provided with a first flow path, a second flow path, and a third flow path, the first flow path extends from a first inlet connectable to an external compressor device for generating compressed air to the first outlet, the second flow path extends from the second inlet to a second outlet for taking out the puncture repair liquid from the extraction cap to supply to a punctured tire to be repaired, the second inlet communicates with the first outlet via the space of the container, the third flow path communicates with the first flow path and the second flow path outside the space of the container so that a portion of the compressed air in the first flow path flows into the second flow path, the first outlet and the second inlet communicate with the space when the internal cap is moved into the space of the container by the compressed air supplied to the first inlet, the extraction cap comprises a closure disposed in the first flow path between the first inlet and the first outlet and configured to close the first flow path between the first inlet and the first outlet, and the first inlet and the first outlet communicate with each other when the closure is moved in a first direction toward the first outlet to pass the first outlet by the compressed air supplied to the first inlet.